# Who Bears the Trade Shocks?
# Distributional Incidence and Fiscal Cushioning in the UK*

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**Abstract**

This paper examines the household incidence of five UK trade events of 2021 to 2026 (the EU customs border, the 2022 energy price surge, US tariffs, and the India and CPTPP agreements) and the extent to which the tax and benefit system cushioned them. The analysis is a policy simulation: the magnitude of each event is imported from published estimates and traced to households, and the two largest are run through the statutory tax and benefit system on representative household microdata covering the full UK population in PolicyEngine UK. Three findings emerge. The negative price events were regressive, with the largest budget-share losses in lower-income deciles. The statutory system provided no automatic offset to price rises, and benefit uprating lagged inflation by one year at a cost of £12 to 15 billion; discretionary support returned 50 to 77 per cent of the energy loss, whereas the post-Brexit food price rise attracted no dedicated instrument. Approximately ninety per cent of the variance in exposure arises within income deciles rather than between them.

**Keywords:** trade shocks, tariffs, Brexit, terms of trade, microsimulation, automatic stabilisers, public finances

**JEL codes:** F13, F14, F16, D31, H23, I38

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# 1 Introduction

Britain has just lived through the densest sequence of trade shocks in its post-war history: three acts of trade policy, a new customs border with its largest trading partner (2021), a tariff war with its largest single-country export market (2025), and a new trade agreement with India (2026), plus the sharpest trade-*transmitted* shock since the 1970s, the 2022 terms-of-trade deterioration, and one near-zero agreement (CPTPP, 2024) that serves as a benchmark. The policy/market distinction is carried explicitly throughout: the energy episode is not a policy act, and the comparison never treats it as one. Each episode has its own literature, and several have been studied at household level, the TCA's food-price effect by income decile, the energy shock's incidence and relief in detail. What is missing is the join: none has been put through the statutory tax–benefit system that determines what households finally keep, and none has been compared with the others under a common lens. This paper does both.

The design holds the second stage fixed and varies the shock. Each episode enters as a *declared first stage*, published estimates for the negative shocks, the government's own projections for the positive ones, never re-estimated here, and is mapped onto households through two channels: consumer prices against expenditure patterns, and labour income through the statutory tax–benefit system. Because gross sizes span two orders of magnitude, comparisons run in ratios: burden as a share of spending by income decile, cushioning shares, and per-pound-of-shock intensities. And because the rulebook itself moved across the window, the Covid uplift removed in 2021, upratings lagging a year behind prices, the Energy Price Guarantee legislated mid-shock, the design treats the tax–benefit vintage as an explicit dimension: the same shock is scored under the rules of its day, under the previous year's rules, and with the discretionary layer switched off.

The first-pass results already discipline the received wisdom in three ways. First, the negative shocks were regressive and the positive episodes were not progressive: on its realised path the energy shock took 5.93 per cent of bottom-decile spending against 2.76 per cent at the top, and the counterfactual it displaced would have taken 8.95 against 4.17, the TCA food-price effect 0.35 against 0.20 per cent on the window-average path, while the India agreement's consumer gains are proportional and worth pounds-per-year, not per cent. Second, each episode's regressivity is generated by its category's budget-share gradient: food takes 14.9 per cent of bottom-decile spending against 8.3 at the top, energy 7.0 against 3.2, clothing and footwear roughly equal shares everywhere, composition, not size, decides who bears a trade shock. Third, the finding the rulebook axis exists for, the British state cushioned its trade shocks with different instruments on different margins: the earnings-led tariff shock was absorbed automatically (36.6 to 43.9 per cent, from the companion study), the price-led energy shock was absorbed by discretion (the Energy Price Guarantee took 33.8 per cent of the counterfactual shock on the financial-year basis of Section 7), and the automatic price-side mechanism, indexation, ran a year late, costing a Universal Credit household £393 (9.8 per cent of the allowance) in the crisis year, months after a £1,040 uplift had been withdrawn.

The contribution is threefold. First, comparability, though the case for it has to be made rather than asserted. That no one has put multiple trade shocks through a common statutory household lens is not in itself an argument for doing so: a gap in a literature can mean the question is hard and worth answering, or merely that the comparison buys little. The defence here is what the join delivers and neither parent literature can. The trade literature measures shocks without the system that redistributes them; the microsimulation literature measures the system against income shocks, for which it is designed. Put together, they show that the statutory system's automatic response to a *price* shock is nil, nominal incomes do not move, so nothing triggers, while the discretionary response was large and progressive, and that the exposure the system is asked to cushion varies far more within income deciles than between them (Section 7). None of those three facts is visible from either literature alone, and each has a policy counterpart. Where the comparison buys less is also worth saying: it cannot rank the episodes on welfare, and its cross-episode ratios are descriptive. Second, the rulebook axis: holding a shock fixed and varying the vintage of the welfare state that receives it has, to our knowledge, no precedent. It is executed here for the largest episode, the same households scored under the 2022 and 2023 parameter systems, which prices the crisis-year uprating lag through the full statutory structure (Section 6.3),and specified for the rest (Section 8); post-Covid Britain, four episodes landing on four rulebooks, is its natural laboratory. Third, honesty about epistemic status as a design principle: every first stage is labelled estimate or projection, every dial is declared, and one episode (CPTPP) is carried as a near-zero benchmark whose only defensible output, an aggregate divided by a household count, is reported as exactly that.

The first pass runs the declared first stages against published expenditure-by-decile tables and statutory arithmetic (Sections 5–6); the two largest episodes, the energy shock and the TCA food shock, are then run through the statutory system on enhanced-FRS household records in PolicyEngine UK (Section 7), which is where the automatic-versus-discretionary decomposition, the rulebook-vintage run and the within-decile dispersion are computed. Section 3 defines the estimand and states what does and does not require identification, Section 4 the five episodes and their declared first stages, Section 8 what extending the household stage to the remaining episodes requires, and Section 9 concludes.

# 2 Literature

Four literatures meet here without previously having met each other: the household incidence of trade policy, the microsimulation of shocks through statutory tax–benefit systems, the distributional analysis of price shocks and benefit indexation, and the UK's episode-specific empirical record since 2020. Each has done part of what this paper does; none has joined them.

## 2.1 Trade to households

The framework for pushing trade shocks onto household microdata is best developed for developing countries: the World Bank's Household Impacts of Tariffs programme simulates tariff changes through both expenditure and income portfolios for 54 countries (Artuç et al., 2021, 2019), and the OECD has published the concordance machinery for mapping CGE trade-model output onto household-budget-survey categories (Luu et al., 2020). The structural anchor for the consumption side is Fajgelbaum and Khandelwal (2016): trade's gains flow disproportionately to lower-income households through traded-goods budget shares. The 2025 tariff war produced a fast-moving US applied literature, the Budget Lab's rolling distributional scoring of the enacted schedules and the Tax Policy Center's implementation of tariffs inside a working tax microsimulation model (Budget Lab, 2025; TPC, 2025), and a first academic treatment of two-sided incidence on household expenditure microdata (Fleck and Pradhan, 2026). The CGE–microsimulation linkage tradition supplies the architecture for putting a trade model upstairs and microdata downstairs (Robilliard and Robinson, 2005; Peichl, 2008; Colombo, 2010; Calì et al., 2019); of that tradition, only Peichl (2008) carries actual statutory tax rules in the micro layer.

What none of this work contains is a tax–benefit system responding to the shock. The World Bank and OECD frameworks stop at market incomes and expenditure; the US 2025 work is consumption-side; the CGE-linkage papers (Peichl's flat-tax application aside) use reduced-form income equations. As far as we can establish, no shock arriving under a *trade* label has been run through a statutory tax–benefit calculator, EUROMOD, UKMOD, TAXBEN, PolicyEngine or any counterpart, other than the companion study discussed below and Section 7 of this paper, which carries both the energy episode and the TCA food-price episode, a trade-policy act, through the statutory stage. The claim is deliberately narrow: price and energy shocks as such have been run through these models (the close antecedents are discussed in the next subsection), and what is new is the trade first stage and the episode-comparative design, not the machinery.

## 2.2 Shocks through statutory rules

That machinery exists and has been applied to every kind of shock except trade. Dolls et al. (2012) measure the stabilisation coefficients of the US and European systems against stylised income and unemployment shocks; Brewer and Tasseva (2021) impose the observed Covid labour-market shock on FRS microdata through UKMOD and decompose the cushioning into automatic and discretionary parts; the EUROMOD nowcasting tradition supplies the labour-transition machinery (Navicke et al., 2014). Two studies come close enough to this paper's rulebook axis that the claim must be stated narrowly. Amores et al. (2025) run the 2022 inflation shock and its offsetting fiscal measures through EUROMOD for the EU, separating income-support from price-containing instruments, in effect a price-shock stabilisation coefficient, and the direct antecedent of what this paper computes; they find fiscal measures recovered roughly a third of welfare losses while targeting them poorly. Popova et al. (2026) are closer still on the UK: UKMOD with LCFS consumption imputed onto the FRS, disposable and consumable income, and an explicit CPI-versus-earnings indexation counterfactual over 2019–2023. Neither models the September-CPI *timing* lag, the Energy Price Guarantee, or any trade shock, and neither treats automatic stabilisation in anything but the classic income-shock sense. That quartet, timing lag, discretionary energy layer, trade first stages, and the comparison across episodes, is what this paper adds, and it is a narrower claim than "nobody has measured price-shock stabilisation."

The conceptual gap it rests on is old and explicit: simulating an oil-price supply shock in FRB/US, Cohen and Follette (2000) found that automatic stabilisers do very little against a supply shock, unlike their demand-shock performance, a result never taken to microdata. Paulus et al. (2020) supply the canonical treatment of non-indexation as a distributional phenomenon, and Leventi et al. (2024) the nearest existing treatment of indexation itself as a stabiliser. The companion study (Ahmadi, 2026) is, to our knowledge, the first tax–benefit microsimulation of a trade shock anywhere, a single episode (the 2025 US tariffs), earnings channel only. This paper generalises it: five episodes, both channels, and the rulebook vintage as an explicit dimension.

## 2.3 Price shocks, indexation and the rulebook

The paper's rulebook axis has more precedent than its trade axis, and the debts should be stated plainly. Cribb et al. (2023) quantified the 2022–23 uprating shortfall directly, benefits rose 3.1 per cent against roughly ten per cent inflation, recipients were not made whole until April 2025, and scored the flat cost-of-living payments against a timely-indexation counterfactual, finding the discretionary route cost more while leaving hundreds of thousands of means-tested households worse off than indexation would have. OBR (2022b) puts the aggregate real fall at around five per cent, some £12 billion, in 2022–23. Judge and Murphy (2023) supply the fact that pre-empts the obvious objection that lags wash out over a cycle: across fifteen years, working-age benefits were uprated below outturn inflation in ten fiscal years, so the asymmetry is empirical, not assumed. Internationally, Eckerstorfer et al. (2025) microsimulate Austria's inflation episode with lagged wage and pension adjustment, the closest non-UK analogue. This paper adds neither the discovery of the lag nor a better estimate of it; what it adds is the join, the same lag priced against *trade-shock* incidence, so that the timing of indexation and the import content of a household's basket are evaluated in one place, across episodes rather than one.

One contrary claim must be met rather than cited in passing. Jaravel (2021) concludes that trade affects price levels roughly similarly across income groups, innovation rather than trade driving inflation inequality. The episodes here are consistent with that as a statement about trade *in general* and inconsistent with it as a statement about these shocks in particular: what makes the UK's negative episodes regressive is not trade exposure as such but the categories they landed on, food and domestic energy, whose budget shares fall steeply with income (Figure 3). Where a trade shock lands on clothing, as the India agreement does, this paper finds exactly the flat incidence Jaravel describes.

The method template for the consumption channel predates all of this: Cravino and Levchenko (2017) showed after Mexico's 1994 devaluation that lower-income households' cost of living rose about half again as much as richer households', because they spend more on tradeables and buy cheaper varieties within categories, the second mechanism being one that category-level data cannot see. The decomposition machinery for price shocks is also established: Sologon et al. (2025) decompose inflation's welfare and distributional impact across expenditure patterns, and Chen et al. (2024) map an exchange-rate shock onto heterogeneous household price experiences using scanner data. Their finding is a direct bound on this paper's first pass: inflation inequality surged in 2021–23 partly because cheaper goods inflated fastest, a within-category quality margin that published expenditure-by-decile tables hold fixed and therefore understate. On the policy question the episodes raise, the sharpest numbers are Italian: Curci et al. (2025) find targeted social bonuses costing a fifth as much as untargeted VAT and excise cuts drove most of the inequality reduction, with nearly half of untargeted energy support accruing above the median.

On the price-response side, Levell et al. (2025) is the frontier and a standing challenge to any decile-based exercise: using transaction data they find income explains only a small share of the variation in energy spending, so within-decile dispersion in exposure dwarfs the between-decile gradient, and they show that an optimal package retains a strictly positive price subsidy even when transfers can condition on income and past usage. That is a principled defence of a price-capping instrument, and it disciplines what this paper's proportional-cushioning arithmetic can be taken to mean. Bonnet et al. (2025) reach the opposite conclusion for transport fuel in France, and the disagreement is commodity-driven, fuel budget shares rise with income where domestic-energy shares fall, which is precisely the kind of composition effect the present framework is built to make visible. Auclert et al. (2023) add the externality a domestic incidence exercise cannot see: national price subsidies insulate one importer at the cost of world energy demand. Cross-country, Ari et al. (2022) and Amaglobeli et al. (2026) document that most European crisis support was untargeted and price-distorting, while NAO (2024) records the fiscal counterpart in the UK, an initial £139 billion estimate against a £44 billion outturn, the option value of a cap that binds only in bad states.

### 2.4 The UK episode record

Each episode carries its own first-stage literature, reviewed with the imported numbers in Section 4: the Brexit/TCA price and trade effects (Bakker et al., 2026; Breinlich et al., 2022; Freeman et al., 2022), the 2022 terms-of-trade and energy-price shock and its policy response, the 2025 US tariff schedule (OBR, 2025), and the India agreement's impact assessment (DBT, 2025). The Brexit episode has the richest distributional record, and two precedents sit close to this paper. Breinlich et al. (2016) asked literally who would bear the pain, distributing projected Brexit price changes across income groups and household types, and Clarke et al. (2017) did the same for an MFN tariff scenario, finding the largest hits on unemployed, single-parent and pensioner households. Both are ex-ante scenario exercises predating the actual border; this paper's contribution relative to them is realised episodes, a statutory second stage, and comparison across shocks. Davenport and Levell (2021) supply the earnings-side counterpart (barrier exposure by occupation, sector and region), and Alabrese et al. (2026) and Dhingra et al. (2026) the 2026 regional accounting, the latter with new TCA tariff and non-tariff-measure estimates in a spatial general-equilibrium model, finding the damage runs through real wages and productivity rather than structural change.

A near-namesake deserves flagging: Pallotti et al. (2023) ask who bore the costs of the euro-area inflation shock and find consumption losses roughly flat across the income distribution, with the incidence falling instead along an age gradient as rigid rents hedged lower-income households. That the same question yields a vertical answer in Britain and an age-based one in the euro area is itself a reason to hold the lens fixed and vary the shock.

The record also contains a puzzle this paper is built to address. Breinlich et al. (2022) find the depreciation episode's consumer cost roughly *flat* across the income distribution; Bakker et al. (2026) find the non-tariff-barrier episode sharply *regressive*, the bottom decile's cost-of-living increase around half again larger than the top's. Same country, adjacent years, opposite incidence, because the two shocks landed on different baskets. A framework that holds the household lens fixed and varies the shock is the instrument that settles such disagreements, and Section 6 shows the same mechanism operating across five episodes. What remains absent everywhere is the tax–benefit response: the Brexit price literature never asks what the benefit system gave back; the energy-crisis literature scores the discretionary response but not the trade shock as such; the FTA impact assessments contain, in DBT's own regulator's words, household analysis "developed further for future FTA IAs." The gap this paper is built to fill is the join: one statutory second stage run against every episode's first stage. This paper supplies the first stages, the taxonomy, a public-data first pass for all five episodes, and the statutory stage for the largest of them; extending that stage to the remaining four is what would complete the argument.

## 3 Framework: one estimand, five episodes, two channels, one moving rulebook

### 3.1 The estimand

For every episode the object is the same conditional quantity: the statutory household response to a declared trade-shock first stage. Let a first stage specify (i) a vector of consumer-price changes by consumption category, and/or (ii) a labour-income loss or gain with a sectoral location and an adjustment margin. The second stage maps that first stage onto households and computes the change in disposable income, its distribution, and the Exchequer's share, under the tax–benefit rules in force at the episode date. The headline metric is the *cushioning rate*, one minus the household disposable-income change per pound of gross shock, the income-stabilisation coefficient of Dolls et al. (2012) generalised to price shocks by measuring the gross shock as the expenditure-weighted cost imposed at fixed consumption, because ratios, unlike levels, are comparable across episodes whose gross sizes span two orders of magnitude.

Nothing in the design estimates the first stages. Each is imported from published estimates and labelled with its source and its attribution caveats (Section 4); the paper's contribution begins where those estimates stop, at the household and the Exchequer. This is the conditional-estimand convention of the companion tariff study (Ahmadi, 2026), applied retrospectively, with the advantage that retrospective first stages can be anchored to realised outturns rather than ex-ante bridges.

### 3.2 Two channels

Trade shocks reach households through prices and through earnings, and the post-Covid episodes split cleanly on that line. The consumption channel prices a category-level price vector against household expenditure shares: implemented at the household level on imputed consumption, done for the energy episode in Section 7, and, for the other episodes, on published expenditure-by-decile tables. The earnings channel imposes sectoral labour-income changes on employees by industry and runs them through the statutory system, with the adjustment margin (wage cuts versus displacement) as a declared scenario dimension, the machinery, conventions and caveats of the companion study carry over unchanged. The channels are reported separately and never summed without an explicit interaction note: a price shock erodes real income that the benefit system partially indexes, while an earnings shock triggers means-tested entitlement; the two responses run through different statutory mechanisms, and their comparison is the point.

### 3.3 The rulebook axis

The same shock cushions differently depending on when it lands, because the rulebook moves: benefit uprating follows lagged inflation, so a fast price shock outruns indexation for a year; thresholds frozen in cash terms tighten in real terms through an inflation episode; and discretionary interventions (most dramatically the Energy Price Guarantee) rewrite the effective schedule mid-shock. The design therefore separates three quantities for each episode: the cushioning delivered by the rules *as they stood*, the cushioning the same rules would have delivered under contemporaneous full indexation (the uprating-lag cost), and the discretionary layer on top. Price-shock incidence has been run through tax–benefit models before (Section 2), and indexation counterfactuals are standard in that literature; what has not been done is to hold a *trade* shock fixed and vary the rulebook vintage that receives it, or to separate the statutory timing lag from the discretionary layer within a single episode. The post-Covid sequence, four episodes landing on four different vintages of the same welfare state, is a natural laboratory for exactly that comparison.

### 3.4 Identification: where the burden sits, and where it does not

Because the paper reports household effects of named policy episodes, it is worth stating precisely which of its numbers require an identifying assumption and which do not. The design deliberately separates the two.

*The second stage requires no identification.* Given a price vector and a household's observed characteristics, what the tax–benefit system pays that household is a computation on statutory rules, not an estimate of a causal parameter. That the means-tested system returns nothing when nominal incomes do not move, that the Energy Price Guarantee returns the difference between two capped prices, that a flat payment is worth more relative to a small bill, these are properties of the rulebook, verifiable line by line, with no counterfactual inference involved. The same is true of the variance decomposition: it describes the dispersion of a computed quantity across observed households. This is the sense in which the exercise is a calculator rather than an estimator, and it is why the second stage can be run at all.

*The first stages carry all the identification, and it is borrowed.* Whether the Trade and Cooperation Agreement *caused* the food-price rise is identified, or not, by Bakker et al. (2026), whose cross-product design and its limits the paper inherits wholesale. Whether the 2025 tariffs will cause the earnings loss the companion study imposes is not identified by anyone; it is declared. The government's projections for the India agreement are model output, not estimates. The design's purpose is to quarantine this problem: each first stage is labelled with its epistemic status (Section 4), and no result in this paper is stronger than the first stage it conditions on.

*One attribution is accounting, not inference.* The claim that part of the 2022 energy shock is trade-transmitted rests on Ofgem's published decomposition of the price cap into wholesale and non-wholesale components, a statement about the composition of a regulated price, not a causal estimate of what world gas markets did to British households.

What follows is that the paper's results should be read as *conditional statutory incidence*: given that a shock of a stated size and shape occurred, this is how the rulebook distributed it. They do not identify the causal effect of Brexit, the tariffs or the agreements on household welfare, and any sentence in this paper that reads that way should be read as shorthand for the conditional statement. The title's question is answered in that sense: not who was harmed by trade policy, but who bore a declared shock once the state had finished responding.

### 3.5 What a decile lens can and cannot see

Two recent results bound what any distribution-by-decile exercise, including this paper's first pass, is entitled to claim. Borusyak and Jaravel (2024) show that for trade shocks the overwhelming share of the variance of household welfare changes lies *within* income deciles rather than between them, because exposure is driven by product-level consumption patterns that income predicts weakly; Levell et al. (2025) reach the same conclusion for energy on UK transaction data. The implication is not that decile gradients are wrong, they are real, and they are what Section 6 measures, but that they are a small part of the dispersion in who was hurt. Decile means answer "was this shock regressive?"; they do not answer "who was hurt?", and the two questions have different answers. This is the strongest single argument for the household-record stage of Section 7, where exposure and entitlement are observed jointly at the household level, and it is why the first-pass results are reported as gradients rather than as welfare statements.

A second bound concerns channels. Decomposing inflation's household impact, Ferreira et al. (2026) find the net-nominal-position and labour-income channels an order of magnitude larger than the consumption-basket channel. The basket channel is the entire content of the price-side exercise here: savers and borrowers, mortgagors and depositors, are not in this paper's ledger, and the earnings channel enters only for the one episode where the companion study supplies it. The estimates are therefore a lower bound on the distributional consequences of these episodes, and are labelled as such throughout.

### 3.6 What the design is not

The estimates are first-round statutory incidence, not general equilibrium: no behavioural response, no wage or employment feedback from the price shocks, no substitution in consumption (the price-channel numbers are first-order, envelope-theorem approximations to welfare changes and upper bounds on avoidable cost). Episode first stages carry their own literatures' uncertainty, which the design brackets by importing low/central/high variants where the literature provides them and labelling single-point stages as such. The comparison across episodes is a comparison of conditional scenarios that share a second stage, not a causal decomposition of realised household outcomes over 2021–2026, which would require separating the trade shocks from everything else that happened, a task this design deliberately declines.

## 4 Five episodes and their declared first stages

Each episode enters the analysis as a declared first stage: published estimates, imported with citations and attribution caveats, never re-estimated. The five span the design's axes: negative and positive, price-led and earnings-led, measured and projected, policy-made and trade-transmitted. One is carried as a near-zero benchmark. Figure 1 places the set on the paper's two axes and Table 1 lists each declared first stage with its source and epistemic status.

### 4.1 Episode 1: the TCA border (2021–23), negative, price-led

The Trade and Cooperation Agreement took effect in January 2021, replacing single-market membership with a zero-tariff but non-tariff-barrier-laden border. The price evidence is the cleanest of any episode: exploiting cross-product variation in EU import intensity, Bakker et al. (2026) attribute roughly 8 percentage points of the 25 per cent food price rise between December 2019 and March 2023 to Brexit non-tariff barriers, a cumulative £250 per household on food alone, £6.95 billion in aggregate (their household count; £250 on the ONS count used here gives £7.1 billion), with pass-through of NTB costs to consumers of 50–80 per cent and the largest effects in high-EU-import products such as meat and dairy (around 10 points above low-exposure comparators). The trade-volume record is equally stark: relative UK imports from the EU fell about 25 per cent on entry (Freeman et al., 2022), and by 2024 goods exports were an estimated £27 billion (6.4 per cent) below counterfactual, with some 16,400 small firms exiting EU exporting entirely (Freeman et al., 2024). The earnings side, by contrast, rests on assumptions rather than estimates, the OBR's standing 15 per cent trade-intensity and 4 per cent productivity wedges (OBR, 2026), so this episode's earnings channel is declared as an assumption-based scenario and its price channel as an estimate-based one. Attribution caveat: the price identification is relative (cross-product), so economy-wide effects may be missed, and the counterfactual is contested by critics of the OBR's trade assumptions.

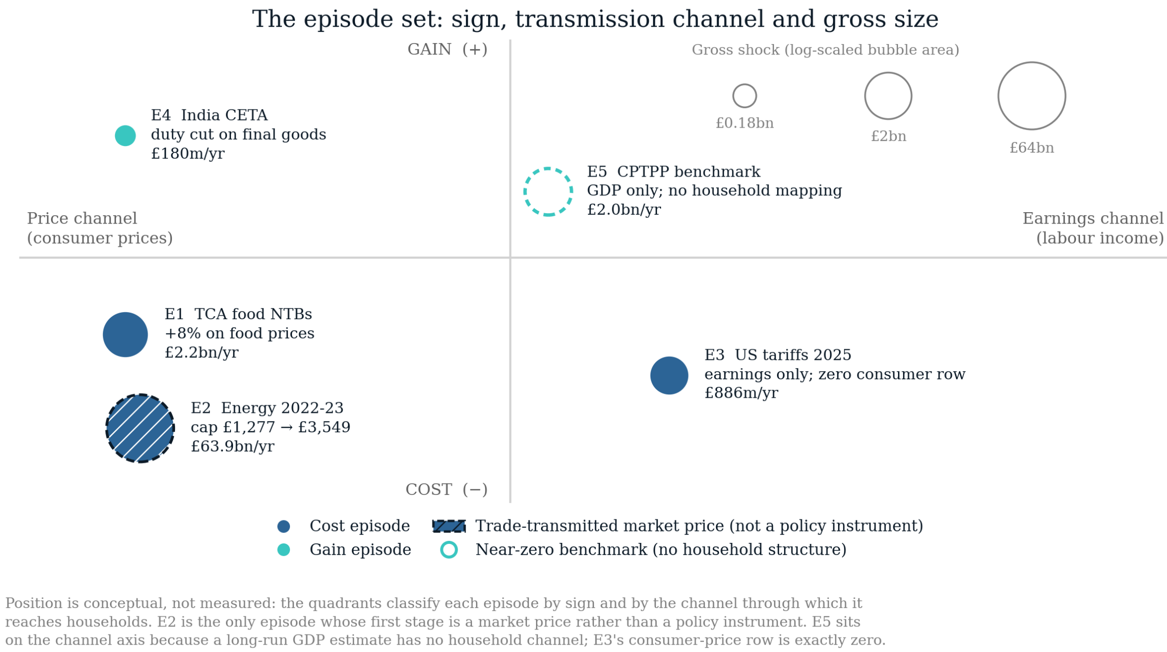



Figure 1: The episode set. Horizontal: the channel through which the shock reaches households; vertical: cost or gain. Bubble area is log-scaled in the gross shock. The 2022 energy episode (hatched, dashed ring) is trade-*transmitted* rather than a policy act; CPTPP (open, dashed) is the near-zero benchmark.

Table 1: Declared first stages. Every price or earnings vector is an imported published estimate; none is re-estimated in this paper.

| Episode | Date | Type | Channel | First-stage source | Key imported number | Status |
|---|---|---|---|---|---|---|
| E1 TCA food NTBs | Dec 2019–Mar 2023$^{w}$ | Policy | Consumer prices (food) | Bakker et al. (2026) | +8% on food prices (6% low variant) | Estimate (ex post) |
| E2 Energy 2022–23 | Oct 2022–Mar 2023 | Market | Consumer prices (energy) | Ofgem (2021, 2022); OBR (2022) | Cap £1,277 → £3,549; EPG £2,500 | Observed (statutory) |
| E3 US tariffs 2025 | 2025 | Policy (foreign) | Earnings only; no UK retaliation | Ahmadi (2026) | £886m/yr gross earnings shock | Estimate (projection) |
| E4 India CETA | 2026+ | Policy | Consumer prices (clothing, footwear, food) | DBT (2025) | £180m/yr duty cut on final goods | Projection (ex ante) |
| E5 CPTPP benchmark | Long run | Policy | Aggregate GDP; no household mapping | DBT (2023) | +£2.0bn GDP (+0.08%) | Projection (long run) |

*Notes.* Type distinguishes a policy instrument from a trade-transmitted market price. Source entries are short cite keys; see the bibliography. E3's consumer-price row is exactly zero (the UK imposed no retaliatory tariffs), so its household incidence runs through earnings alone. $^{w}$ the *estimation* window of the cited study; the TCA itself entered into force in January 2021.

## 4.2 Episode 2: the terms-of-trade shock (2022–23), negative, price-led, very large

The 2022 energy and import-price surge was, in the Bank of England's description, a terms-of-trade shock of a size unseen since the mid-1970s: the terms of trade deteriorated 6.2 per cent in the year to 2022Q3 (ONS, 2023), import prices rose 20 per cent relative to output prices over two years (Broadbent, 2022), the energy import bill roughly doubled to £117 billion in 2022 (IMF, 2023), the *increase* of roughly £60 billion being a transfer abroad of about 2.5 per cent of GDP (derived arithmetic, labelled as such), and of the roughly 17 per cent CPI rise to 2022Q4, more than 70 per cent (12.7 points) was imported (Dhingra and Page, 2023). The household-level price vector is policy-mediated and well documented: the typical annual energy bill's cap rose from about £1,277 (winter 2021–22) to £3,549 (October 2022), truncated to £2,500 by the Energy Price Guarantee, at a discretionary cost the OBR put at £27 billion for the EPG within £47 billion of price subsidies and £78 billion of total energy support (OBR, 2022). Counterfactual no-policy welfare losses average 6 per cent of after-tax income, reaching 11 per cent at the 95th percentile of exposure (Levell et al., 2025), against energy budget shares of roughly 11 per cent in the second income decile versus 6 per cent in the ninth (Resolution Foundation, 2022). This is the strongest first stage of the five, multiple independent official estimates of both the aggregate shock and the retail price vector, plus a published imported/domestic decomposition, and the design decision it forces is constitutive for this paper: the shock is declared *gross*, and the Energy Price Guarantee is modelled as what it was, a discretionary rulebook intervention, so that automatic and discretionary cushioning can be separated.

## 4.3 Episode 3: the US tariffs (2025–26), negative, earnings-led

The April 2025 US schedule (10 per cent baseline, 25 per cent autos and steel) and its partial mitigation are the subject of the companion study (Ahmadi, 2026), which supplies this episode's earnings first stage and its second-stage results: a tariff-calibrated earnings loss of roughly £886 million a year, cushioned at 43.9 per cent when delivered as broad wage cuts and 36.6 per cent as displacement. The external anchors are the OBR's March 2025 scenarios (import-demand elasticity 0.4; peak GDP effects of $-0.6$ to $-1.0$ per cent across scenarios) (OBR, 2025) and the Bank of England's effective-tariff estimates (11 per cent at April 2025, about 9 per cent after the Economic Prosperity Deal) (Bank of England, 2025). The UK consumer side of this episode is, to date, literally zero: the UK has not retaliated, so no tariff-induced consumer price vector exists, an empirical fact the comparative design records rather than an omission.

## 4.4 Episode 4: the India agreement (2026–), positive, both channels

The UK–India Comprehensive Economic and Trade Agreement entered into force on 15 July 2026. Its first stage is declared from the government's own impact assessment (DBT, 2025): long-run GDP $+£4.8$ billion a year (+0.13 per cent by 2040), real wages +0.19 per cent ($\approx £2.2$ billion), and, on a different clock, at entry into force, consumer duty reductions of £220 million a year, £180 million of it on final goods, concentrated in clothing (£132 million), footwear (£13 million) and food and beverages (£12 million). Sectoral GVA effects give the earnings channel its geography: machinery and equipment +1.65 per cent, textiles and apparel $-0.68$ per cent. Two dials are declared rather than estimated, following the IA's own caveats: preference utilisation (the duty figures assume 100 per cent on static 2022 trade) and consumer pass-through of the duty savings. The honest framing, and the reason a government projection can anchor a conditional design, is that the exercise computes what the government's own central estimates imply for households, which its impact assessment does not report and its regulatory scrutineer asked for (RPC, 2026).

### 4.5 Episode 5: CPTPP accession (2024–), the near-zero benchmark

The UK's accession to the CPTPP (in force December 2024) carries a central government estimate of +£2.0 billion a year in the long run, 0.08 per cent of GDP, about £70 per household per year before any incidence structure, because the UK already had agreements with nine of the eleven members (DBT, 2023). It is included precisely because it is nearly nothing: a data-limitation benchmark (the replication artifacts label it accordingly) whose honest treatment, an aggregate divided by a household count, and no more, marks the boundary of what a declared first stage can support.

### 4.6 What the five episodes jointly reveal about the record

One pattern runs across the dossier and should be stated carefully, because its naive version is mechanical: past shocks naturally have ex-post estimates and new agreements ex-ante projections. The non-mechanical fact is on the evaluation side, no UK trade agreement has ever been ex-post evaluated at the household level, although CPTPP has now been in force long enough for the exercise, and the India assessment's own regulatory scrutineer flagged the consumer analysis as underdeveloped (RPC, 2026). The comparative design keeps the epistemic split explicit throughout, estimate-based and projection-based first stages are labelled in every table, and the episode set is typed policy/market, with the terms-of-trade episode carried as a trade-transmitted market shock, never as a policy act.

## 5 First-pass method

This section describes the first pass, which runs the declared first stages against public aggregates for all five episodes; the energy episode is additionally run on household records in Section 7, whose method is set out there. Everything is generated by one pipeline script (`incidence.py`) that downloads and pins its raw inputs and emits every reported value as a machine-generated macro; no number is typed by hand except the declared first-stage constants of Section 4, which are typed *with their citations* and marked as imported.

Following the companion studies' convention, the epistemic status of every input is stated once:

| Object | Status |
|---|---|
| Episode first stages | **Imported** from cited published estimates (episodes 1–2) or government projections (episodes 4–5); episode 3 from the companion study. Never re-estimated here. |
| Expenditure by category × decile | **Data** (ONS Family Spending detailed workbooks). |
| Energy price-cap levels, statutory benefit rates, CPI outturns | **Data** (Ofgem/DESNZ, DWP, ONS). |
| Pass-through and utilisation dials | **Assumed**, bracketed, entering the pipeline once each. |
| Incidence arithmetic | **Computed**; no estimation step anywhere in the pipeline. |

**Consumption incidence.** Each episode's price vector is applied to the relevant categories of household expenditure by equivalised-disposable-income decile, giving a per-decile burden or gain in pounds a year, as a share of total spending, and as a share of disposable income. These are first-order calculations: no substitution, no behavioural response, fixed baskets, upper bounds on avoidable welfare cost, in the envelope-theorem sense of Deaton (1989), whose first-order incidence method this literature, from Fajgelbaum and Khandelwal (2016) onward, descends from. Where a published aggregate exists, its role differs by episode and is stated in the results: the OBR's Energy Price Guarantee cost is a context anchor beside the pipeline's own aggregate, while the TCA's £250 published cumulative is used as an *anchor*, the pipeline's linear-ramp cumulative overshoots it by a factor of 1.68, and the headline TCA path is rescaled to the published figure (Section 6), so that episode's level is the published estimate's, not the pipeline's.

**Rulebook arithmetic.** The vintage axis is seeded with statutory arithmetic on public parameters: the real-terms shortfall of the Universal Credit standard allowance during the 2022–23 inflation episode relative to contemporaneous indexation (the uprating-lag cost, computed month by month against CPI outturns), and the discrete removal of the £20-a-week uplift in October 2021, the rulebook event that immediately preceded the terms-of-trade shock. The Energy Price Guarantee enters as the discretionary layer of episode 2: the gap between the gross price shock and the EPG-truncated shock is the discretionary cushioning share, computable per decile from cap levels alone.

**Two data caveats.** The FYE2025 Family Spending workbook carries an ONS correction notice affecting percentage standard errors across the high-level COICOP categories; the point estimates this paper uses are unaffected, but no sampling-uncertainty statement is made on the expenditure inputs for that reason. And the two vintages used (FYE2022 for the TCA and energy episodes, whose spending bases must predate the shocks, and FYE2025 for the India agreement) sit either side of an ONS change in the equivalence scale, so income levels are not comparable across vintages, which is why every cross-episode comparison in this paper runs on shares of spending rather than shares of income.

**Comparability.** Episodes are compared in ratios, not levels: burden as a share of spending by decile, cushioning shares where computable, and a per-£-billion-of-shock normalisation, because gross sizes span more than two orders of magnitude and pound totals would reduce the comparison to a statement about the energy episode. Scenario dials (pass-through, utilisation) are reported as labelled variants, never blended into central cells; brackets are outer bounds, not confidence intervals; and estimate-based versus projection-based first stages are flagged in every table.

# 6 First-pass results

## 6.1 The comparative anatomy

Table 2 is the paper's central object: the five episodes side by side under the common lens. Three regularities organise everything that follows. First, *the negative shocks were regressive and the positive episodes were not progressive*: on the realised 2022–23 path the energy shock cost the bottom decile 5.93 per cent of its spending against 2.76 per cent at the top (the never-realised gross counterfactual would have been 8.95 against 4.17), the TCA food shock 0.35 against 0.20 per cent of spending on the window-average path (1.19 against 0.66 at end-state), while the India agreement's consumer gains are spread almost exactly proportionally (0.0151 versus 0.0154 per cent of spending) and are two orders of magnitude smaller. Second, *regressivity follows the affected category's budget-share gradient*: food takes 14.9 per cent of bottom-decile spending against 8.3 at the top, energy 7.0 against 3.2, while clothing and footwear, where the CETA gains land, are flat across the distribution: the composition of a shock, not just its size, determines who bears it. Third, *the cushioning came from different parts of the state*: the one earnings-led episode was cushioned automatically (36.6–43.9 per cent through the tax–benefit system, from the companion study), while the price-led energy episode was cushioned by discretion, the Energy Price Guarantee absorbed 33.8 per cent of the gross shock on the financial-year basis (the point-to-point cap ratio is 46.2 per cent), and the automatic price-side mechanism, benefit indexation, arrived a year late (Section 6.3).

## 6.2 Episode detail

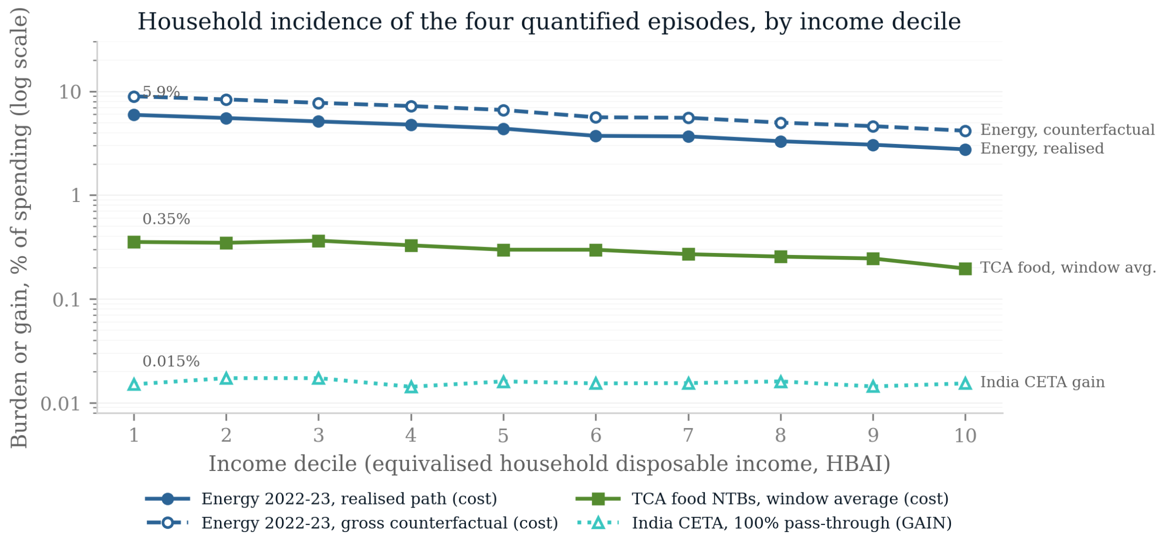
**TCA food prices.** The headline incidence uses the window-average path implied by the published cumulative: the £250-per-household estimate implies an average effect of about 0.30 of the 8 per cent end-state over the window, giving £58 (bottom decile) to £94 (top decile) a year, £2.2 billion a year in aggregate, regressive as a share of spending (0.35 versus 0.20 per cent) because food takes 14.9 per cent of bottom-decile spending against 8.3 at the top. (The estimation window opens in December 2019, thirteen months before the TCA took effect; the linear-ramp and window-average conventions both absorb the pre-entry months as zeros.) This is an anchoring choice, not a cross-check that happened to pass: the pipeline's own linear-ramp cumulative overshoots the published figure by a factor of 1.68, and the window-average path is obtained by rescaling to the published cumulative. The TCA row therefore carries no information from the pipeline beyond the ONS food-share gradient, and the level is the published estimate's. The end-state figures (£194–315 a year; £7.37 billion) are the unrescaled variant: the pipeline's linear-ramp cross-check against the published cumulative comes in at ratio 1.68, showing the price effect was back-loaded, so end-state overstates the lived burden. The household stage would carry the published path if this episode were extended to it (Section 8).

**The energy shock and the EPG.** What households actually faced in 2022–23 was a mean capped bill of £2,236 (£1,971 for the first half-year, the Energy Price Guarantee's £2,500 for the second) against £1,208 the year before: a realised rise of 85.1 per cent costing the bottom

Table 2: Five shocks under one lens: first-pass comparability

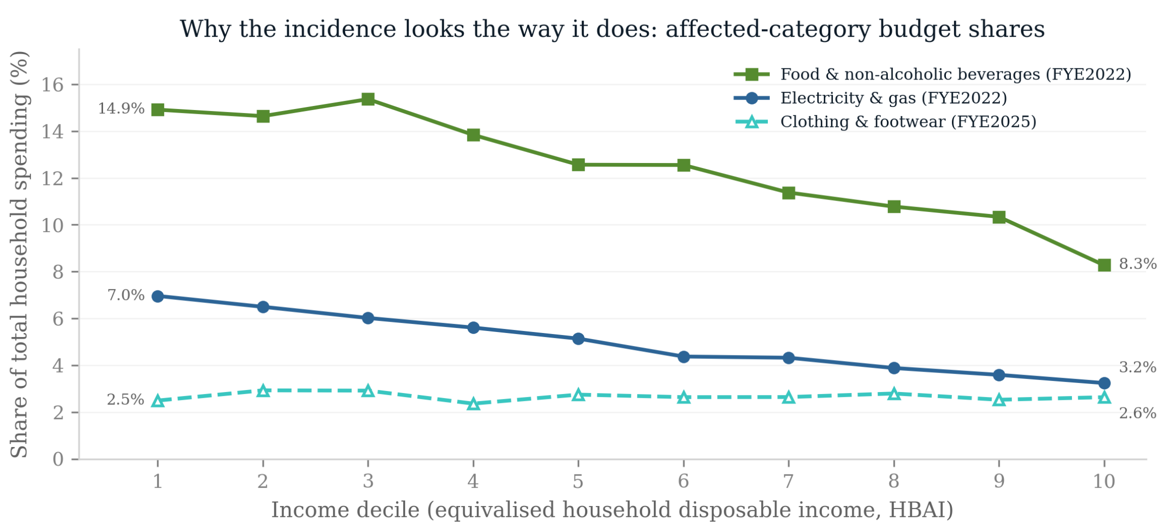
| Episode | Type, channel | Shock (realised path) | % of spending D1 | % of spending D10 | Cushioning, by mechanism |
|---|---|---|---|---|---|
| TCA food, 2021–23 | policy, prices | £2.2 bn | 0.35 | 0.20 | none legislated (0%)$^{t}$ |
| Energy, 2022–23 | market, prices | £30.6 bn | 5.93 | 2.76 | price subsidy, 33.8%$^{i}$ |
| US tariffs, 2025–26 | policy, earnings | £886 m | zero | | , $^{m}$ |
| India CETA, 2026– | policy, prices | £180 m$^{g}$ | 0.0151 | 0.0154 | , |
| CPTPP, 2024– | benchmark | £2.0 bn$^{g}$ | no structure | | , |

*Notes:* $^{g}$ gain, not cost. $^{t}$ the common cushioning estimand, computed at the household stage (Section 7.7): no discretionary instrument was legislated against this episode and the automatic response to a price shock is structurally zero. $^{i}$ an accounting identity: because the Guarantee capped a price, the share is constant across deciles by construction; it is on the financial-year basis used throughout (the point-to-point cap ratio is 46.2 per cent). $^{m}$ this column carries only the common cushioning estimand of Section 3, computed at the household stage for the two price episodes; the tariff episode's earnings-channel stabilisation coefficient (36.6–43.9 per cent, imported from the companion study (Ahmadi, 2026)) answers a different question and is reported in this note rather than in the column. Deciles are of equivalised disposable income; entries are first-order incidence on published expenditure-by-decile tables at fixed baskets. Both price episodes are on their *realised* paths, but on different clocks, and the asymmetry matters for the D1 comparison: energy is an annualised crisis-year rate, while TCA is the published cumulative averaged over a window padded with pre-entry zeros, at the end-state (crisis-period) rate the TCA row reads 1.19 against 0.66 per cent. TCA is on the window-average path implied by the published cumulative, energy on the caps households actually faced in 2022–23 (£2,236 against £1,208, +85.1 per cent). The end-state and gross-counterfactual variants are reported in the text. Episodes are typed policy/market: the energy episode is trade-*transmitted*, not a policy act. "Zero" records that the UK did not retaliate, so no consumer price vector exists. Cushioning entries measure different mechanisms (a pass-through convention, a price subsidy, a tax–benefit stabilisation coefficient) and are not summable; the common cushioning estimand of Section 3 is computed at the microsimulation stage, not here. Category budget-share gradients, which generate these incidence patterns, are in Figure 3; full per-decile detail in Table 3 (whose derived columns are computed by the figure generator from the pinned artifact, on the same FY basis).




Costs are drawn with solid or dashed lines and filled markers; the CETA gain is dotted with open markers and labelled GAIN. All four series are plotted as magnitudes, so the log axis compares sizes, not signs. Energy is the October 2022 cap shock; TCA is the window-average path (0.30 of the end-state +8% effect).

Figure 2: Household incidence by equivalised disposable income decile, as a share of each decile's total spending. Costs are plotted as positive; the India CETA series is a gain (dotted, open markers). The energy episode is shown on its realised 2022–23 path (solid) and as the gross counterfactual the Energy Price Guarantee displaced (dashed).




Food and energy shares fall steeply with income, which is what makes those two episodes regressive; the clothing & footwear share is flat, which is what makes the CETA gain roughly proportional. Each share is computed on its own episode's spending vintage (ONS Family Spending 3.1E), so levels are not strictly comparable across the two vintages.

Figure 3: Why each episode has the incidence it does: the affected categories' shares of total spending by decile. Food and domestic energy fall steeply with income; clothing and footwear, where the India CETA gains land, are flat.

Table 3: Household incidence of the quantified episodes, by equivalised disposable income decile. TCA food NTBs are on the window-average path (0.30 of the end-state +8% effect); energy is the 2022–23 cap shock, gross and net of the Energy Price Guarantee; the India CETA column is the consumer gain at 100% pass-through. Costs are positive.

| | TCA food NTBs | | Energy, gross | | Energy, net of EPG | | CETA gain |
|---|---|---|---|---|---|---|---|
| Decile | £/yr | % spend | £/yr | % spend | £/yr | % spend | £/yr |
| 1 | 58 | 0.35 | 1,458 | 8.95 | 965 | 5.93 | 2.96 |
| 2 | 63 | 0.35 | 1,504 | 8.35 | 996 | 5.53 | 3.93 |
| 3 | 70 | 0.37 | 1,478 | 7.74 | 978 | 5.12 | 4.47 |
| 4 | 74 | 0.33 | 1,631 | 7.21 | 1,080 | 4.78 | 3.90 |
| 5 | 72 | 0.30 | 1,591 | 6.61 | 1,054 | 4.38 | 5.12 |
| 6 | 84 | 0.30 | 1,584 | 5.62 | 1,049 | 3.72 | 5.37 |
| 7 | 80 | 0.27 | 1,651 | 5.56 | 1,094 | 3.68 | 5.61 |
| 8 | 84 | 0.26 | 1,631 | 4.99 | 1,080 | 3.31 | 6.56 |
| 9 | 90 | 0.25 | 1,698 | 4.62 | 1,124 | 3.06 | 6.90 |
| 10 | 94 | 0.20 | 1,986 | 4.17 | 1,315 | 2.76 | 9.93 |
| D1/D10 ratio | — | 1.80 | — | 2.15 | — | 2.15 | 0.98† |

*Notes.* £ per household per year; % spend is the burden or gain as a percentage of that decile's total expenditure. The EPG cushions 33.8% of the gross energy shock at every decile by construction. † CETA ratio is on the %-of-spending basis (£ gains are not comparable across deciles). Spending bases: ONS Family Spending workbook 1, sheet 3.1E, FYE2022 vintage (TCA, energy) and FYE2025 vintage (CETA).

decile £965 a year, 5.93 per cent of its total spending against 2.76 per cent at the top, and £30.6 billion in aggregate. The *counterfactual* shock, the October 2022 cap of £3,549 carried through the financial-year mean, giving £2,760 against the FY2021–22 mean of £1,208, +128.6 per cent, is the figure the policy prevented, and would have cost the bottom decile £1,458 a year, 8.95 per cent of its spending. Three labelled qualifications bound that number, and each is a different kind of bound. First, *welfare versus cash*: the fixed-basket figure is the first-order welfare measure, correct to a first approximation by the envelope theorem, but households cut consumption, so cash outlays rose by less. Applying the observed weather-adjusted reduction in domestic gas consumption of 13 per cent over the gas year to mid-May 2023 (DESNZ, 2024) gives a cash-outlay burden of 77 per cent of the fixed-basket figure (£1,128 for the bottom decile; £35.8 billion aggregate). The two answer different questions, what households paid, and what they lost, and the gap between them is itself the foregone consumption. Second, *how much of this is trade*: the retail cap bundles wholesale energy, which is traded, with network, policy and operating costs, margin and VAT, which are domestic. Ofgem's published component breakdown settles the split (Ofgem, 2021, 2022). Wholesale rose from £550 to £2,468 of the typical bill, from 43.1 to 69.5 per cent of the cap, so 84.4 per cent of the cap's *rise* is traded energy and the remainder domestic cost. The trade-transmitted component of the realised burden is therefore about £25.8 billion (£39.0 billion of the gross counterfactual). One clarification matters for what this number means. Wholesale is not the same as imported: the wholesale price is set on a world-linked market, so UK-produced gas repriced too, and the resulting transfer runs from British households to British producers and, through the Energy Profits Levy, the Exchequer, not abroad. The wholesale share therefore measures exposure to a world-market-linked price rather than a transfer overseas, and the trade-transmitted figures above should be read in that narrower sense. The comparison with Dhingra and Page (2023)'s finding that *more than* 70 per cent of the CPI rise was imported is offered only as an order-of-magnitude check, since their statistic concerns the aggregate price index rather than the wholesale component of one regulated tariff. Third, *timing*: the annualised run-rate holds the October 2022 cap for a full year, which no household experienced; the six-month window figure is £7.8 billion of EPG absorption against the OBR's £27 billion costing, a context anchor, not a validation. Within those bounds the shape is robust: the Energy Price Guarantee truncated the rise to +85.1 per cent, absorbing £492 a year for a bottom-decile household, 33.8 per cent of the gross shock on the financial-year basis (£16.5 of £48.7 billion, Section 7); the point-to-point cap ratio, (3,549-2,500)/(3,549-1,277), is 46.2 per cent and is not used further. Because the EPG capped the *price*, its cushioning share is constant across deciles by construction. Two qualifications follow from the frontier evidence. First, decile means understate the problem the instrument faced: Levell et al. (2025) show that income explains only a small share of the variation in household energy spending, so exposure within a decile is far more dispersed than the gradient between deciles, an argument for the household-record stage, where that dispersion is observable, and against reading any decile-mean cushioning share as a statement about individual households. Second, a proportional instrument is not thereby a mistaken one: Levell et al. (2025) find that an optimal package retains a strictly positive price subsidy even when transfers can condition on income and past usage, precisely because income targets energy exposure so poorly. The finding here is descriptive, the cap cushioned proportionally, not a welfare verdict. Whether the overall discretionary response was progressive cannot be judged from the EPG alone: the Energy Bills Support Scheme's flat £400, the means-tested cost-of-living payments and the council-tax rebate, jointly of the same order as the EPG and deliberately targeted, are not modelled in this first pass. Section 7 settles the question on household records: with the whole stack modelled, the discretionary response was progressive, not proportional.

One comparator needs reconciling. Resolution Foundation (2022) report energy budget shares of roughly 11 per cent in the second decile against 6 per cent in the ninth; the shares computed here are 6.5 and 3.6 per cent. The gap is definitional, not substantive: this paper measures electricity and gas only, as a share of *total spending*, on equivalised-income deciles of the FYE2022 vintage, whereas the wider comparator includes other domestic fuels and is expressed against income. Because the episode's incidence is linear in this share, the difference scales the level of every energy number reported here, another reason the household-record stage, which computes the share directly rather than importing it, supersedes this pass.

**The US tariffs.** The UK consumer side is zero, no retaliation occurred, so this episode's incidence is entirely earnings-led: a gross loss of £886 million a year cushioned by the automatic tax–benefit system at 36.6 per cent (displacement) to 43.9 per cent (wage cuts), the companion study's headline carried over as this episode's second stage.

**The India gains, and the near-zero benchmark.** Of the £180 million of final-goods duty reductions, £157.2 million falls in the three categories with published spending detail (clothing, footwear, food and beverages) and is allocated across deciles; the remainder is unmodelled. At full pass-through the CETA duty cuts are worth £2.96 a year to a bottom-decile household and £9.93 to a top-decile one, roughly proportional as a share of spending, though larger in pounds at the top (the top decile gains about three times as much in cash), and economically negligible at any pass-through dial (£1.48–4.97 at 50 per cent). The CPTPP row is a near-zero benchmark, included as a data-limitation case rather than a placebo test: its £2.0 billion

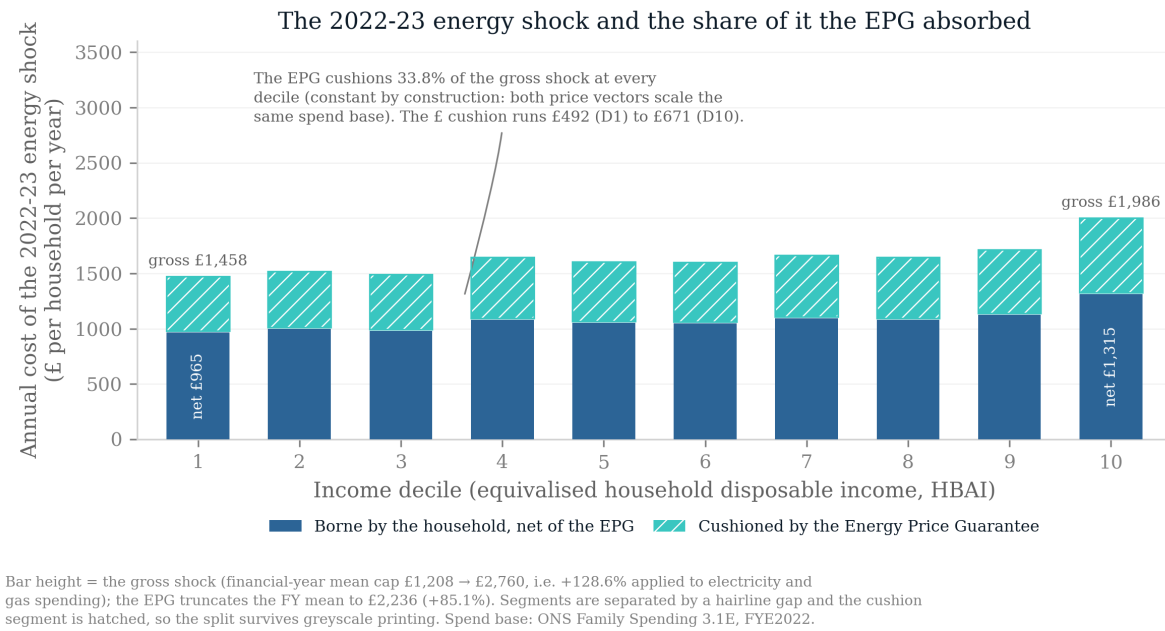



Figure 4: The 2022–23 energy shock by decile: the gross cap-level burden split into the part borne by households net of the Energy Price Guarantee and the part the Guarantee absorbed. Because the instrument capped a price, its cushioning share is constant across the distribution by construction.

long-run GDP estimate divides to £70 per household per year with no defensible distributional structure, and the pipeline reports exactly that and nothing more.

## 6.3 The rulebook seed results

The vintage axis is seeded with two statutory facts. First, the uprating lag: Universal Credit's standard allowance rose 3.1 per cent in April 2022 (indexed to the previous September's inflation) while prices rose at double digits. Against month-by-month contemporaneous indexation, an upper-bound counterfactual no benefit system operates, a single adult's standard allowance fell £393 short over 2022–23 (9.8 per cent of the annual allowance); against the policy-feasible alternative of a single April 2022 uprating at contemporaneous annual CPI, £230. Both figures are for the single-adult standard allowance only, before elements and caseload weighting; the range, not either point, is the seed result. Neither the lag nor its cost is a discovery of this paper: Cribb et al. (2023) quantified the 2022–23 shortfall and scored the discretionary cost-of-living payments against a timely-indexation counterfactual, and OBR (2022b) put the aggregate real fall at roughly £12 billion, a figure the second stage's £18.0 billion (Section 7) does not reproduce: the bases and price counterfactuals differ, the two are stated side by side rather than reconciled, and the reconciliation is an explicit task of the extension (Section 8). What is new here is the join: the lag priced against trade-shock incidence, on the same households, across episodes. The lag also does not wash out over the cycle: Judge and Murphy (2023) count ten of fifteen years in which working-age benefits rose by less than outturn inflation. The shortfall is lost to timing alone, at exactly the moment the terms-of-trade shock peaked. Figure 5 shows

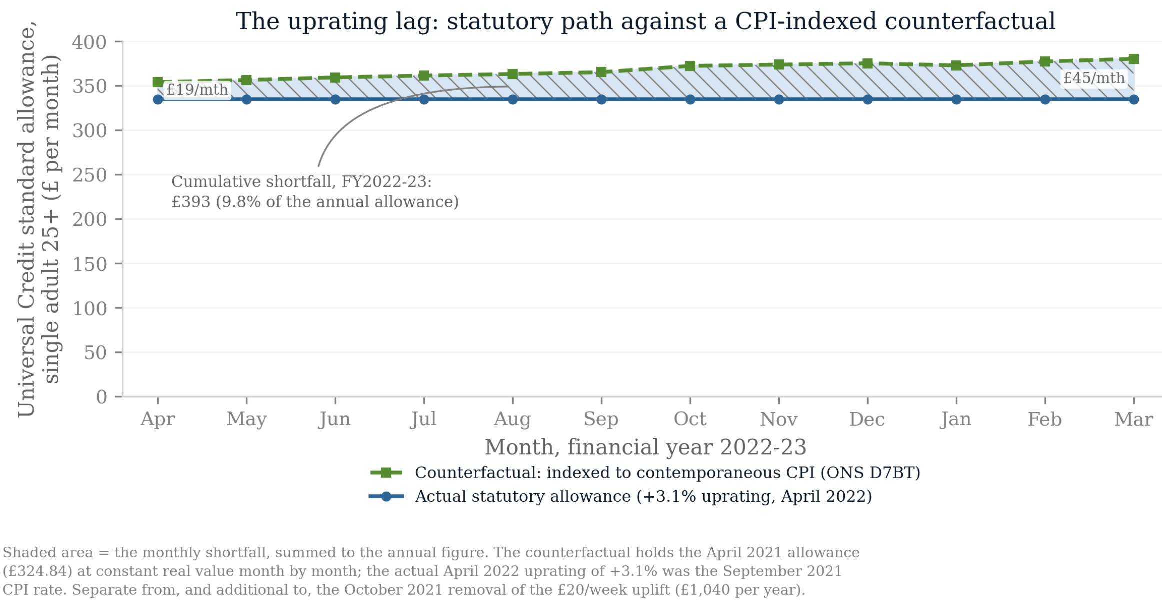



Figure 5: The uprating lag in the crisis year. The Universal Credit standard allowance for a single adult aged 25 or over (statutory path, flat from April 2022) against a counterfactual indexed to contemporaneous CPI. The shaded area is the shortfall, summing to £393 over 2022–23.

the gap month by month. Second, sequencing: the £1,040-a-year Covid uplift was removed in October 2021, months before the energy shock landed, so the lowest-income households entered the largest price-led trade shock of the period from a rulebook trough. Together with the EPG's absorption of 33.8 per cent of the gross price shock, the seed picture of the British response is: *automatic on the earnings margin, discretionary on the price margin, and lagged precisely where the price shocks bit hardest*, with the progressivity of the full discretionary stack an open question for the microsimulation stage. The microsimulation stage turns these representative-household facts into population-weighted decompositions.

# 7 The second stage: the energy episode on households

## 7.1 Method

The largest episode is run through the statutory system on 53,508 enhanced-FRS household records in PolicyEngine UK 2.89.2 (dataset `enhanced_frs_2023_24.h5`, 2023 vintage), in the tax–benefit microsimulation tradition surveyed by Figari et al. (2015). Four conventions govern the numbers that follow and each is a choice a reader could make differently.

*One basis for both paths.* The counterfactual and realised energy paths are both financial-year mean caps: the FY2022–23 counterfactual is £2,760 (the April and October 2022 caps, no Guarantee), the realised path £2,236 (April 2022 cap then the Guarantee), and the FY2021–22 baseline £1,208. This matters: comparing a point-to-point cap ratio with a financial-year mean, as is tempting, silently changes the denominator of the cushioning rate.

*Rebasing (assumption).* The enhanced FRS calibrates energy spending to current-vintage price levels, so its level reflects today's prices while the cross-household distribution reflects quantities. The weighted mean is rebased to the FY2021–22 mean cap, a factor of 0.606, preserving each household's relative position. Every shock aggregate below is linear in this factor and should be read as conditional on it; the within-decile results are invariant to it. The cushioning *rates* are not: the flat-cash instruments (the Bills Support Scheme, the cost-of-living payments) do not scale with the rebase while the shock denominator does, so the discretionary rate and the decile cushioning shares depend on the factor. The dependence is quantified in the two-axis grid of Table 4.

*Coverage.* The Energy Price Guarantee enters as the difference between the two capped paths, the Energy Bills Support Scheme as its flat £400 to every weighted household (so the modelled £12.6 billion exceeds the roughly £11 billion scheme outturn by about 14 per cent: the 10.5 per cent weight excess plus a small residual from non-domestic-supply households in the weighted base), and the 2022–23 cost-of-living payments on simulated entitlement in the simulation year, an approximation in both directions: simulated entitlement is not receipt in the actual 2022 qualifying windows, the £150 disability payment is modelled per household rather than per qualifying individual, and Housing-Benefit-only households are included although they did not qualify for the £650. The council-tax rebate and the Warm Home Discount are not modelled. The discretionary total is therefore an approximation with omissions on both sides, not a bound.

*Rule year (assumption).* The simulation scores the episode on the 2023 parameter system and 2023 incomes, the vintage of the enhanced-FRS base, although the episode is FY2022–23. The rulebook-vintage run below scores the same households under the 2022 parameter system as the crisis-year counterfactual; the discretionary-stack results are insensitive to the rule year because the instruments enter as declared cash amounts, and the cost-of-living eligibility flags are nearly identical under the two vintages (8.1 against 8.2 million flagged households).

*Weights.* All weighted arithmetic is delegated to the `microdf` survey-weighting library that PolicyEngine itself returns results in; no weight calculation is hand-rolled. The person weights gross to 69.8 million people, in line with the UK population. PolicyEngine's household grossing weights total 31.4 million households against an ONS count of 28.4 million, so aggregates carry an upward bias of about 10.5 per cent; ratios are unaffected. Deciles are equivalised on the HBAI before-housing-costs concept, and burdens are expressed as shares of household expenditure rather than income, because a ratio with income in the denominator is dominated by very low incomes. One consequence must be flagged: the burden *numerator* is rebased to FY2021–22 cap levels while the expenditure *denominator* stays at the enhanced FRS's current-vintage price level, and that base is larger than the ONS FYE2022 tables the first pass divides by; second-stage shares of spending are therefore roughly half their first-pass counterparts by construction, and the two stages' shares are comparable in shape but not in level. The variance decomposition and the within-decile quantile ratios are unaffected: putting the denominator on the same price base rescales every household's share by a common factor, which changes neither the between/within split nor any ratio of quantiles. Aggregate pounds, which share a basis, reconcile to within about five per cent.

*Reconciliation with the first pass.* On the same financial-year basis the first pass puts the counterfactual shock at £46.2 billion and the realised path at £30.6 billion, against £48.7 and £32.3 billion here. The residual gap of about five per cent is smaller than the 10.5 per cent excess of PolicyEngine's grossing weights over the ONS household count, so the household base explains at most part of it: the imputed consumption distribution and the rebase convention offset the remainder, and the decomposition of the gap is not attempted here. Ratios are unaffected and levels should be read with the unreconciled gap in mind.

### 7.2 The automatic response, measured

Every means-tested award in the UK conditions on nominal income, and a price shock leaves nominal income untouched: the automatic response is therefore zero by the structure of the statutory rules, not as an empirical finding. The tax–benefit calculator is run twice, once on baseline inputs and once with energy inputs shocked to the realised path, and the change in every entitlement is £0.00 billion (0.0 per cent of the shock); but because no modelled entitlement takes energy consumption as an input, this re-run confirms that the model encodes the statutory structure rather than measuring an independent quantity, and the claim rests on the rulebook, which is verifiable line by line. It is also a first-order claim: a shock of this size moved nominal incomes in general equilibrium, wage bargaining, self-employment margins, and those excluded channels (Section 3) would trigger means-tested responses the first-order calculation cannot. This is Cohen and Follette (2000)'s macro finding, that automatic stabilisers do little against a supply shock, at the household level, and sharper than their version, because for a pure price shock the mechanical response is not merely small but absent.

Two qualifications keep the claim honest. It is a statement about the *modelled* statutory system, so channels outside it, the Warm Home Discount, capital drawdown altering means-tested entitlement, passported eligibility shifting at the margin, self-employed income responses, could move the true figure off zero, though all are small relative to the shock. And it is an in-year statement: indexation is the system's automatic price-side channel, and it arrives late rather than never.

### 7.3 Indexation arrives late

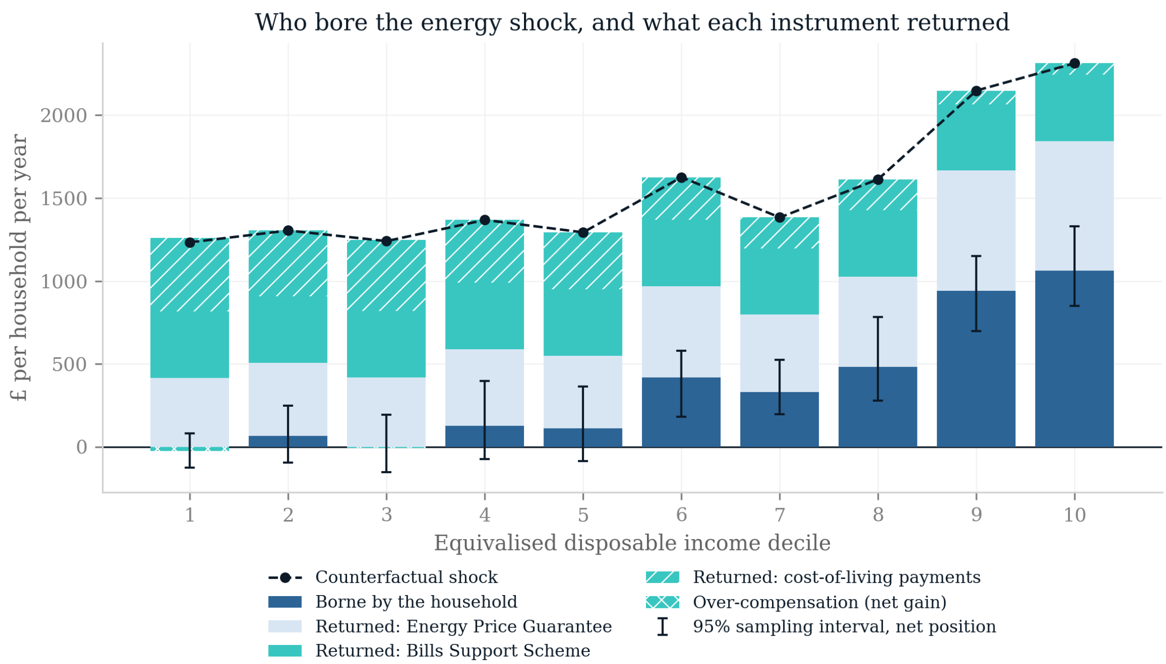
Benefits rose 3.1 per cent in April 2022, indexed to the previous September, against inflation over the year of about 10 per cent, the benefit-erosion mechanism whose cross-national treatment is Paulus et al. (2020). The cost of that gap is here computed with a genuine rulebook-vintage run rather than a flat base-times-gap multiplication: the enhanced FRS is a single-period dataset, so every input column is copied to the 2022 period and the *same* households are scored under the 2022 and 2023 parameter systems. The legislated April 2023 uprating (10.1 per cent) moves the fourteen uprated cash benefits from £235.3 to £256.8 billion,£21.6 billion through the full statutory structure, tapers included, or £2.13 billion per uprating point. At that measured rate, the 7.0 points withheld in the crisis year cost £14.9 billion (the final step is linear in the rate, an approximation declared as such), against the roughly £12 billion aggregate real fall of OBR (2022b) on a different base and counterfactual, and below the £18.0 billion a flat multiplication of the 2023-rules base (£257 billion) would give, because tapers claw back part of any uprating. It is deliberately *not* expressed as a share of the energy shock: the numerator is the erosion of benefit income by all inflation, and dividing it by one commodity's counterfactual would attribute to energy what food, services and rents also did. What the number establishes is narrower and still worth having, that the one automatic channel available to a price shock was running backwards while the shock landed.

## 7.4 The discretionary response was large and progressive

Against a counterfactual shock of £48.7 billion (128.6 per cent on the FY basis), the *energy-specific* instruments, the Guarantee (£16.5 billion) and the Bills Support Scheme (£12.6 billion), returned 59.5 per cent (66.2 per cent at the bottom decile, 51.1 at the top). Adding the cost-of-living payments (£8.7 billion) takes the full stack to £37.7 billion, or 77.4 per cent. The attribution choice must be stated: the cost-of-living payments were legislated against the general inflation episode, not energy specifically, so dividing them by the energy-only counterfactual over-attributes to this shock by the same logic on which the preceding subsection refuses to divide the indexation shortfall by the energy shock; both rates are therefore carried, the energy-specific one as the conservative reading. On either attribution it was progressive, on the full stack, 102 per cent of the bottom decile's counterfactual loss returned against 54 per cent of the top decile's, because flat payments are worth more relative to a small bill and the targeted payments went where they were aimed. The over-compensation of the bottom decile exists only on the full-stack attribution: energy-specific, the bottom decile recovers 66.2 per cent. Two features of the profile deserve stating rather than smoothing. The bottom decile is *over*-compensated on this accounting, at 102 per cent, as is the third at 100; and the gradient is not monotone, the sixth decile (74 per cent) sitting below the fifth. The over-compensation is narrow in two ways: the bootstrap below puts it within sampling error of exact compensation, and it says only that the instruments returned about as much as the counterfactual energy cost to those households, not that they were made whole against inflation generally, which the preceding subsection shows they were not. The first-pass inference that discretion cushioned *proportionally* was an artefact of modelling the price cap alone; adding the instruments the cap sat alongside reverses it, and the correction runs in the government's favour. Figure 6 shows the anatomy in pounds: the counterfactual shock by decile, split into the part the household bore and what each instrument returned. A household bootstrap (999 resamples of the record base, following the practice urged for microsimulation results by Goedemé et al. (2013)) puts a 95 per cent sampling interval of 74.7–80.5 per cent around the 77.4 per cent aggregate rate, and 94.1–111.2 per cent around the bottom decile's 102: the over-compensation of the bottom decile is therefore not statistically distinguishable from exact compensation, and the net borne amount spans zero in each of the bottom five deciles (95 per cent intervals £-125–80, £-96–248, £-152–193, £-72–398 and £-84–364 a year for deciles one to five). These are sampling intervals over the survey base only; imputation uncertainty from the consumption donor models is not captured (Section 7.6).

## 7.5 The two-axis sensitivity grid

Table 4 varies the two conventions the headline rates depend on: the rebase factor (the paper's computed 0.606, an intermediate 0.8, and 1.0, no rebase) and the discretionary stack, built up one instrument at a time. Two facts organise the twelve cells. First, the Guarantee's cushioning rate (33.8 per cent) is invariant to *both* axes, to the rebase because a price cap scales with the bill it truncates, and across deciles by the accounting identity already noted, so every pound of progressivity and every point of rebase-sensitivity in the headline comes from the flat-cash instruments. Second, the headline is bounded, not fragile: the full stack cushions 77.4 per cent of the counterfactual shock at the paper's rebase and 60.2 per cent with no rebase at all, and the progressive gradient survives every cell: without rebasing, the full modelled stack returns 75.2 per cent of the bottom decile's loss against 46.1 per cent of the top decile's. The qualitative findings, discretionary, large, progressive, hold across the grid; the bottom decile's




Source: fig_extra.py on the enhanced FRS (PolicyEngine UK); FY basis; whiskers are household-bootstrap 95% intervals.

Figure 6: The anatomy of the energy episode in pounds a year by equivalised disposable income decile: the counterfactual shock (dashed line), the part borne by the household net of all modelled instruments, and the parts returned by the Energy Price Guarantee, the Bills Support Scheme and the cost-of-living payments. Whiskers are 95 per cent household-bootstrap sampling intervals on the net position; in the bottom five deciles it is indistinguishable from zero. Sampling uncertainty only; imputation uncertainty is not captured.

Table 4: Cushioning under two conventions: rebase factor × discretionary stack

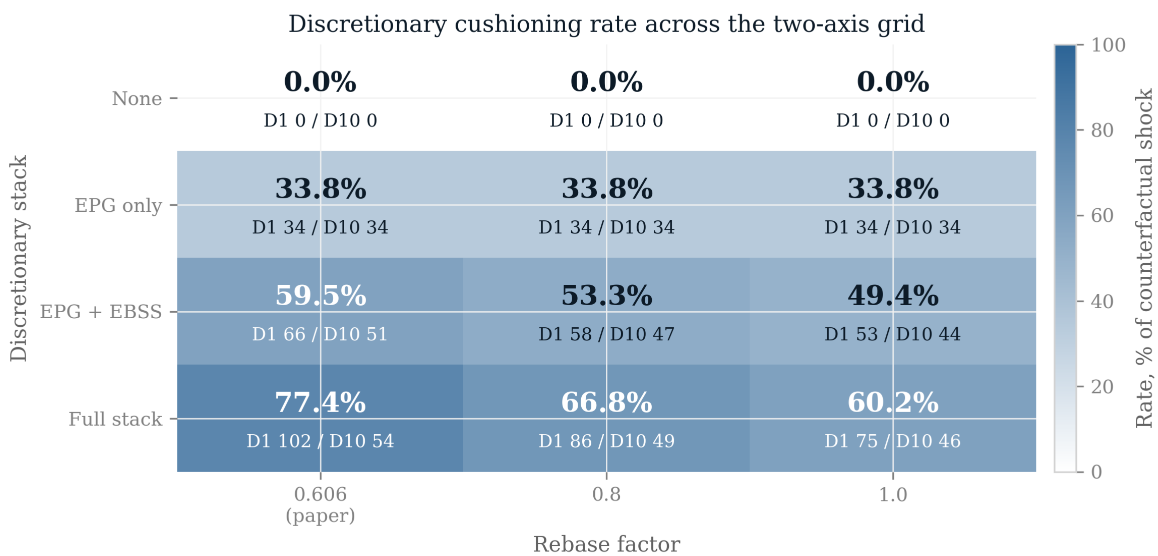
| Rebase | Stack | Shock (£bn) | Rate (%) | D1 (%) | D10 (%) |
|---|---|---|---|---|---|
| 0.606 (paper) | None | 48.7 | 0.0 | 0.0 | 0.0 |
| | EPG only | 48.7 | 33.8 | 33.8 | 33.8 |
| | EPG + EBSS | 48.7 | 59.5 | 66.2 | 51.1 |
| | Full stack | 48.7 | 77.4 | 102.0 | 54.0 |
| 0.800 | None | 64.3 | 0.0 | 0.0 | 0.0 |
| | EPG only | 64.3 | 33.8 | 33.8 | 33.8 |
| | EPG + EBSS | 64.3 | 53.3 | 58.4 | 46.9 |
| | Full stack | 64.3 | 66.8 | 85.5 | 49.1 |
| 1.000 | None | 80.4 | 0.0 | 0.0 | 0.0 |
| | EPG only | 80.4 | 33.8 | 33.8 | 33.8 |
| | EPG + EBSS | 80.4 | 49.4 | 53.4 | 44.3 |
| | Full stack | 80.4 | 60.2 | 75.2 | 46.1 |

*Notes:* Each cell is arithmetic on the same simulated households; the measured automatic response is zero in every cell (Section 7), so no re-simulation is required. "Rate" is the discretionary cushioning rate against the counterfactual shock on the financial-year basis; D1 and D10 are the bottom- and top-decile rates. The EPG row's rate is constant across rebase and deciles by construction.

over-compensation is the one result that is basis-dependent, present at the paper's rebase (102.0 per cent) and absent without rebasing (75.2 per cent). One external anchor bears on the choice of cell and must be confronted rather than footnoted: at a rebase of one the EPG absorption is £27.2 billion, close to the OBR's £27 billion EPG costing, while the paper's rebase gives £16.5 billion. The near-match is partly coincidental, the OBR priced the Guarantee against the higher caps then expected for 2023, over a longer window, on price-dependent volumes, but it cautions against reading the rebased cell as the truth and the unrebased one as a variant. The paper's headline statements are therefore confined to what holds across the grid, and the basis-dependent over-compensation is reported as exactly that. One dial the grid does not vary is the counterfactual *path* itself: holding the October 2022 cap through a full financial year is a convention, not a forecast, the Guarantee pre-empted that cap before it ever bound, and an ex-ante articulated path would give a different denominator. The choice is declared (Section 7.1) and varying it is extension work.

### 7.6 Exposure varies within deciles, not between them

Figure 9 plots the three shock paths of the design on one distribution: the gross counterfactual, the realised path, and the cash-outlay path (the gross counterfactual scaled by the observed demand response, the same convention as the first pass). The paths differ in level, not shape: policy and demand response scaled the shock down without changing who it fell on. The realised burden runs from 2.77 per cent of household expenditure in the bottom decile to 1.35 per cent in the top. The gradient is regressive and not monotone, the third decile (2.78 per cent) sits marginally above the first. The share *levels* are about half their first-pass counterparts by the denominator convention declared in Section 7.1, so only the shape, not the level, is comparable across the two stages; within that comparison, energy exposure separates households far less cleanly by income than decile tables imply. Decomposing the variance of that burden puts 6.6 per cent of it between deciles and 93.4 per cent within them on the paper's preferred basis




Source: grid_energy.py on the enhanced FRS (PolicyEngine UK); rates on the financial-year basis.

Figure 7: The two-axis grid of Table 4 as a heatmap. Cell shading is the discretionary cushioning rate against the counterfactual shock; each cell also reports the bottom- and top-decile rates. The EPG row is invariant to both axes, a price cap scales with the bill it truncates and is constant across deciles by construction, so the progressivity and the rebase-sensitivity of the headline both come entirely from the flat-cash instruments.

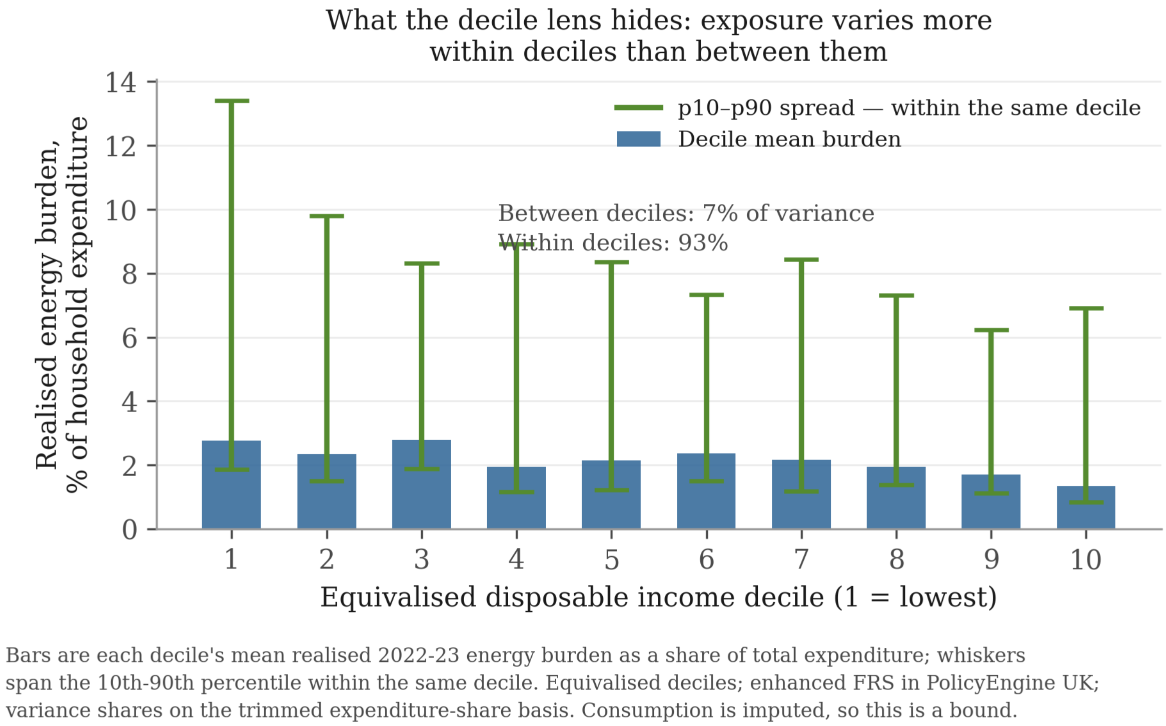
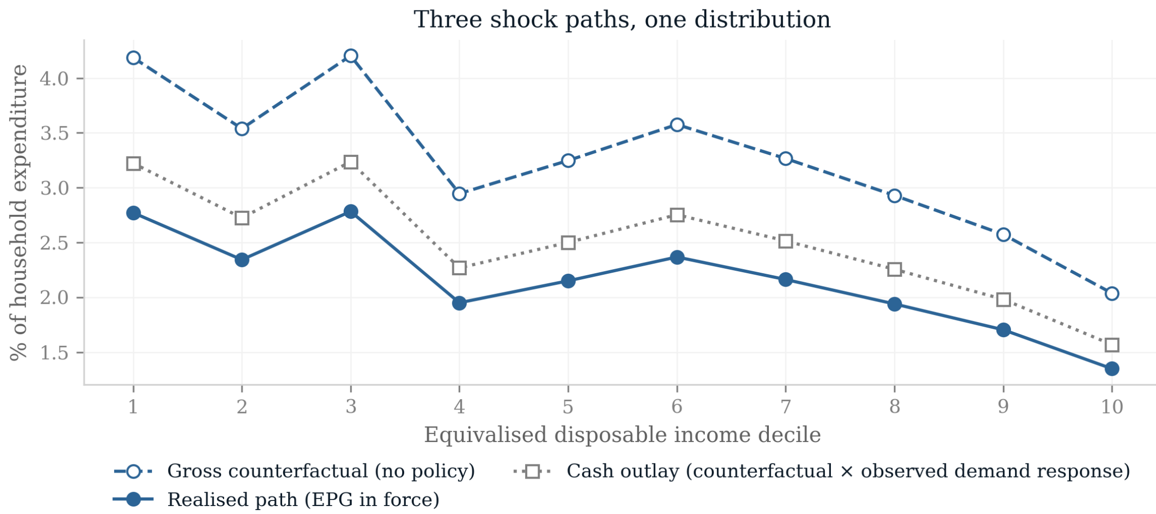



Figure 8: What the decile lens hides. Bars are each decile's mean realised energy burden as a share of household expenditure; whiskers span the tenth to ninetieth percentile of that burden *within* the decile. On the trimmed expenditure-share basis the between-decile gradient accounts for 6.6 per cent of the variance in exposure and the within-decile spread for 93.4 per cent.

(expenditure share, trimmed at the first and ninety-ninth percentiles; household-bootstrap 95 per cent interval 90.1–94.8 per cent for the within-share); untrimmed the between-share is 0.1 per cent and in logs 1.4 per cent. The trimmed expenditure-share figure is the paper's estimate; the near-zero untrimmed between-share is dominated by a handful of extreme imputed shares in the tails and is reported as a data-quality diagnostic on the imputation, not as a robustness check. On every basis, though, the gradient the first pass measured is a small part of the dispersion in who was exposed. Within the bottom decile alone the burden runs from 1.86 per cent of spending at the tenth percentile to 13.40 per cent at the ninetieth.

Two limits bound what this establishes. The exposure is *imputed*, enhanced-FRS energy spending is a quantile-regression forest fitted to LCFS donors, so part of the within-decile dispersion is donor-model variation, and the decomposition is a bound rather than a measurement of true dispersion, the bootstrap interval above is sampling uncertainty only, blind to the donor-model error, and a direct validation of the imputed within-decile spread against the LCFS donor sample, which observes it, is left to the extension; Levell et al. (2025) use transaction data precisely for this reason. And the estimand is the variance of one commodity's expenditure share, not the variance of welfare changes from product-level trade exposure, which is what Borusyak and Jaravel (2024) decompose: the results point the same way, but they are not the same quantity.




Source: fig_extra.py on the enhanced FRS (PolicyEngine UK); shares on the second stage's hybrid basis.

Figure 9: Three shock paths by equivalised disposable income decile, as a share of household expenditure, on the second stage's financial-year basis: the gross counterfactual (no policy), the realised path (Energy Price Guarantee in force), and the cash-outlay path (the gross counterfactual scaled by the observed weather-adjusted demand response, the first-pass convention). Level shifts, common shape.

## 7.7 A second episode through the second stage: the TCA food shock

The common cushioning estimand is computed for a second episode, and this one is a trade-*policy* act. The TCA food-price first stage (the window-average path of Section 6) is priced against each household's imputed food consumption on the same 53,508 records, rebased so the weighted mean matches the first pass's ONS FYE2022 food-spend base (factor 1.096; the same convention as the energy episode's cap rebase, with the same hybrid share denominators). The aggregate, £2.4 billion a year, therefore matches the first pass by construction of the mean, the new content of this stage is the distribution, not the level. The burden is regressive (0.18 per cent of bottom-decile spending against 0.11 at the top, on the second stage's share basis) and larger in pounds at the top (£54 at the bottom against £128 at the top), and 95.9 per cent of the variance in exposure lies within deciles (trimmed basis; within the bottom decile the burden runs 0.12–1.13 per cent of spending at the tenth to ninetieth percentile). The cushioning decomposition is short and is the point: the automatic response is structurally zero, as for any price shock; indexation ran late, as above; and no episode-specific instrument exists, nothing was legislated against the TCA food-price effect. The contrast with the energy episode must be drawn on the episode-specific basis, by the same attribution logic as the discretionary subsection above: the general cost-of-living payments that are the non-conservative part of the energy stack also cushioned food-price inflation in 2022–23, so they cannot count toward energy and as zero for food. Episode-specific, the contrast stands and is the paper's sharpest comparative fact: against one price-led trade shock the state legislated instruments returning 59.5 per cent of the counterfactual loss; against the other, of the same sign and the same regressivity, it legislated none. Composition and salience, not incidence, decided which shock got its own instruments.

## 7.8 From incidence to demand and welfare

The statutory incidence measured above is an accounting object. This subsection aggregates it into two economic ones, keeping the paper's conventions: every behavioural parameter is imported from the literature and declared, nothing is estimated, and the outputs are first-round policy-simulation quantities, no general-equilibrium feedback, no intertemporal response.

*Demand.* Each flow of the energy episode, the counterfactual burden and the three instruments, is weighted by a marginal propensity to consume assigned by equivalised-income decile, under three declared scenarios: a gradient falling from 0.8 in the bottom decile to 0.3 in the top, the income and liquidity pattern of Jappelli and Pistaferri (2014) and Fagereng et al. (2021); a flat 0.5, which switches the heterogeneity channel off; and an asymmetric UK variant after Bunn et al. (2018), who find MPCs out of income *losses* (about 0.5) far larger than out of gains (about 0.14). Table 5 reports the results. Three facts follow. First, the counterfactual shock would have destroyed £25.2 billion of first-round consumption; with the modelled stack in place the net effect is £-4.4 billion, the support protected most of the demand the shock threatened. Second, the instruments rank by targeting: the cost-of-living payments protected 0.62 of consumption per pound of fiscal cost, the flat Bills Support Scheme 0.55, the proportional Guarantee 0.52, the demand-efficiency ordering that MPC heterogeneity predicts. Third, the asymmetric scenario is the sobering one: if households cut spending out of losses at 0.5 but spend transfers at only 0.14, the net first-round demand effect is £-19.1 billion, the stack protected incomes, but much less of demand. The same logic gives the uprating lag its demand-side reading: the £14.9 billion withheld by late indexation was withheld from benefit recipients, the highest-MPC households in the economy, so the demand drag per pound withheld was at its maximum.

Table 5: First-round demand effects of the energy episode under declared MPC scenarios

| | First-round demand effect, £bn | | |
|---|---|---|---|
| Flow | MPC gradient | MPC flat | Fiscal flow |
| Counterfactual burden (no policy) | -25.2 | -24.4 | -48.7 |
| Energy Price Guarantee | 8.5 | 8.2 | 16.5 |
| Bills Support Scheme | 6.9 | 6.3 | 12.6 |
| Cost-of-living payments | 5.4 | 4.3 | 8.7 |
| Net position (full modelled stack) | -4.4 | -5.5 | -11.0 |

*Notes:* MPC gradient: 0.8 (bottom decile) to 0.3 (top), linear; MPC flat: 0.5 everywhere. The asymmetric variant (Bunn et al., 2018: 0.5 out of losses, 0.14 out of gains) gives a net effect of £-19.1 billion (not shown by row). First-round, no general-equilibrium feedback; parameters imported, not estimated.

*Welfare.* Weighting the counterfactual burden by Atkinson (1970) social-welfare weights on equivalised income,the simplest member of the generalised welfare-weight framework of Saez and Stantcheva (2016), (proportional to $y^{-\gamma}$, normalised to mean one) gives a regressivity premium of 0.95 at $\gamma = 1$ and 1.26 at $\gamma = 2$. The $\gamma = 1$ figure is deliberately reported although it is unflattering: the shock is regressive in *shares* but larger in *pounds* at the top, so at log utility the pound-concentration at the top roughly offsets the share-concentration at the bottom, and only at higher inequality aversion does the welfare cost exceed that of an equal-pounds shock. Distributional statements about this episode therefore depend on the inequality aversion the reader brings, and the paper reports the mapping rather than choosing for them.

# 8 Extending the household stage

Section 7 runs the energy episode through PolicyEngine UK on the enhanced Family Resources Survey. Extending the same stage to the remaining four episodes is a parameter reform per episode rather than new machinery, and four things change when it is done.

First, the consumption channel moves from decile tables to household records: the enhanced FRS carries imputed household consumption across the twelve COICOP divisions plus the sub-items that matter most for trade shocks (petrol and diesel separately, in pounds and litres; electricity and gas separately; bus fares), imputed by quantile regression forest from the LCFS 2023–24 and calibrated to administrative energy and fuel totals. Each episode's price vector therefore prices each household's own basket, jointly with that household's benefit entitlements, the covariance between exposure and cushioning that decile tables cannot see. Tariff-line and category-level first stages must be pre-aggregated to those roughly sixteen categories, and the imputation freezes budget-share *patterns* at the 2023–24 donor vintage across simulated years, a limitation the stage will state. Second, the earnings channels run through the companion study's displacement machinery on the same households, so the two channels of a given episode land on a common population and their incidence can be compared household-by-household rather than distribution-by- distribution. Third, the rulebook axis becomes exact: PolicyEngine's parameter history carries dated, legislatively cited values through the whole window (the Covid-era Universal Credit uplift and its October 2021 removal, the March 2022 fuel-duty cut, the 2022 and 2023 upratings), its own test suite exercises the 2020–2023 rule years, and counterfactual-rulebook runs, an earlier year's parameters applied to a later survey vintage, are native to the architecture. Each episode is therefore scored under the rules in force at its date, under the preceding year's rules (isolating uprating lag), and with the discretionary interventions switched off (isolating the automatic component), the three-way decomposition of Section 3 computed rather than approximated. Two verified limits shape the earlier years: no 2021-vintage enhanced-FRS data year exists (the earliest survey base is 2022–23), so 2021 rulebooks are applied to later households with incomes re-deflated; and the survey's Universal Credit/legacy-benefit split follows reported receipt rather than the historical rollout, so 2021–22 rule-year results understate legacy-benefit interactions and are restricted to the tax and UC-parameter channels. Fourth, the poverty and inequality outputs gain their correct household equivalisation and the Exchequer effects their full statutory coverage.

The build order follows the data dependencies. The consumption-channel implementation extends PolicyEngine's existing consumption variables (used today for carbon and VAT analysis) with the episode price vectors, a parameter reform per episode, no new imputation. The earnings channels are re-runs of existing machinery with episode calibrations. The rulebook runs require only parameter-year switches plus explicit flags for the discretionary layers (the Energy Price Guarantee being the largest). The licensed inputs are the same FRS/EFRS files the companion study uses; nothing else is gated.

Two methodological upgrades are available at the same time and are specified here. First, the fixed-weight decile indices of the first pass can be replaced by non-homothetic distributional cost-of-living indices, which Hochmuth et al. (2025) construct from exactly the inputs this project already has, aggregate prices, expenditure shares and a single cross-sectional expenditure distribution, answering the standard objection that fixed baskets misprice large relative-price movements. Second, and more consequentially, the household-record stage is the level at which the within-decile dispersion documented by Borusyak and Jaravel (2024) and Levell et al. (2025) becomes visible: the same machinery that computes a decile gradient computes the full distribution of exposure within it, which is where most of the welfare variance lives.

Two honest limits will survive the full stage. The consumption imputation carries its own error, inherited from the LCFS donor models, and imputation error correlated with benefit status would bias the exposure-cushioning covariance; the stage will report the imputation diagnostics alongside the results. And the rulebook decomposition attributes to "discretion" whatever the parameter history records as legislated change, which bundles genuine shock response with policy that would have happened anyway, a labelling choice, stated as such, not an identification claim.

# 9 Discussion

The household stage settles three things the first pass could only frame. The statutory system's automatic response to a price shock is not small but structurally absent: entitlements condition on nominal income, which a price shock leaves untouched, so the only automatic channel is indexation, and indexation arrived a year late, leaving benefit income 7.0 per cent short in real terms over the year in which the shock landed. Everything that did cushion was discretionary, and once the whole stack is modelled rather than the price cap alone it turns out to have been both large (roughly half to three-fifths of the shock energy-specific across the sensitivity grid; three-fifths to three-quarters with the general cost-of-living payments attributed to this shock) and progressive, a correction that runs in the government's favour and that the decile-table pass had got wrong. The second episode carried to household records sharpens this into the paper's central comparative fact: the TCA food shock, price-led, regressive, a policy act, had no instrument legislated against it, so the margin-specificity of the British response extends to a starker selectivity: which price shock gets a discretionary stack is decided by salience and composition, not by incidence. And the exposure the state was trying to cushion varies overwhelmingly *within* income deciles (93.4 per cent of the variance on the preferred basis), which is why an income-conditioned instrument targets it poorly and why a price instrument retains a defensible role (Levell et al., 2025).

With that in hand, three findings survive every caveat the first pass carries. The negative shocks of the 2020s, two policy-made, one trade-transmitted, were regressive through prices, and the positive policy shocks offer no offsetting progressivity: partly because they are small, partly because their gains arrive in categories (clothing, footwear) whose budget shares are flat across the distribution. The state's cushioning was margin-specific: automatic where the shock hit earnings, discretionary where it hit prices, and the largest discretionary instrument, capping a price, cushioned proportionally by construction, and, once the whole stack is modelled on household records rather than the cap alone, progressive as well (Section 7). Proportional is not the same as mistaken: because income predicts energy exposure poorly, a positive price subsidy survives in the optimal package (Levell et al., 2025), and the cap's fiscal cost was itself state-contingent, an initial £139 billion estimate against a £44 billion outturn (NAO, 2024). The instrument's defensibility is commodity-specific: where budget shares rise with income, as for transport fuel, transfers dominate subsidies (Bonnet et al., 2025); the Guarantee applied to the commodity where the case for a price instrument is strongest. What a domestic incidence exercise cannot price at all is the externality that national subsidies impose through world energy demand (Auclert et al., 2023). And the automatic price-side stabiliser the welfare state does possess, indexation, is calibrated to a world where prices move slowly: against a fast terms-of-trade shock it delivered its help a year late, which for the lowest-income households was the difference between the rulebook cushioning the shock and amplifying it.

One institutional fact deserves stating precisely, because its naive version is a truism: that past shocks have ex-post estimates and recent agreements ex-ante projections is mechanical. What is not mechanical is that the United Kingdom has never ex-post validated a trade agreement's impact assessment at the household level, CPTPP has been in force long enough for the exercise, and its absence is a fact about evaluation practice that the Regulatory Policy Committee's criticism of the India assessment's consumer analysis makes concrete (RPC, 2026). On the government's own projections, the positive policy episodes are small and distributionally flat; on the published estimates, the negative episodes are large and regressive. Whether that ledger balances is exactly the question a household-level evaluation practice would answer, and no one keeps the books.

The limits are those declared throughout. First stages are imported, with their identification and their controversies; the incidence is first-order and static; the channels are not summed; the rulebook decomposition labels legislated change as discretion without asking what would have been legislated anyway. One episode has been carried to household records; the other four have not, and their first-pass numbers should be read accordingly. Poverty and inequality effects, the remaining episodes' rulebook decompositions, and the consumption channel's interaction with the earnings channel on common households all remain for the extension Section 8 specifies.

What the framework offers, finally, is a standing capability rather than a one-off study: every future trade shock, a retaliation schedule, the next agreement, the next terms-of-trade event, arrives as one more declared first stage for the same second stage. A country that intends to conduct an active trade policy while running a means-tested welfare state needs exactly this join; we are not aware of an institution that maintains it.

## Data and code availability

The first-stage evidence dossier, the costing/incidence pipeline, pinned raw downloads and generated result macros are available in the accompanying GitHub repository (`https://github.com/PolicyEngine/uk-trade-shock-study`, directory `analysis_multishock/`); an archived release with a persistent identifier will accompany publication. All inputs to the first-pass results are public (ONS expenditure and price statistics, published first-stage estimates, statutory parameters). The microsimulation stage additionally requires the licensed Family Resources Survey via the UK Data Service, which is not redistributed. Episode first stages are imported from cited published estimates and are not re-estimated here.

## Declarations

The author reports no conflicts of interest. No external funding was received for this study. The project uses secondary, aggregate public statistics and published estimates; no new human-participant data were collected. The author is responsible for the analysis, interpretation, and any remaining errors.

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